# Yin ZHANG

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## EDUCATION

**New York University, USA** **Sep 2026 – May 2028 (Expected)**
*M.S. in Computer Engineering*

**Nanyang Technological University, Singapore** **Aug 2025 – May 2026**
*GEM Trailblazer Exchange Student, B.Eng. in Data Science and Artificial Intelligence*

- **Relevant Modules:** Natural Language Processing, Computer Vision, Human-AI Interaction & Communication

**Tianjin University, China** **Sep 2022 – Jun 2026**
*B.S. in Data Science and Big Data Technology (Minor: Computer Science and Technology)*

- **Relevant Modules:** Data Structures, Algorithm Design and Analysis, Advanced Programming, Methods of Optimisation, Mathematical Analysis, Probability Theory, Mathematical Statistics, Complex & Real Variable Functions.

## RESEARCH INTERESTS

- **Trustworthy AI**: Advancing reasoning faithfulness and robust safety mechanisms in LLMs.
- **Multimodal Safety & Efficiency**: Improving VLM training efficiency and cross-modal safety alignment through information-theoretic reinforcement learning.
- **Mechanistic Interpretability**: Understanding how internal representations, features and circuits give rise to model behavior and developing trustworthy causal interventions for model safety and reasoning.

## PUBLICATIONS & MANUSCRIPTS

[1] **Zhang, Y.***, Wu, Z.*, Zhao, J., Fang, J., Li, Z., Wang, K., Wen, Q., & Shao, Y. (2026). See First, Reason Later: Mutual Information-Guided Reinforcement Learning for Vision-Language Models. *Accepted by International Conference on Machine Learning (ICML) 2026.* [arXiv:2605.01520]

[2] Zhu, X., Zhu, B., Fang, J., Wang, S., **Zhang, Y.**, Wang, X., & He, X. (2025). GuardAlign: Robust Safety Alignment in Multimodal Large Language Models. *Accepted by International Conference on Learning Representations(ICLR) 2026.* [arXiv:2602.24027]

[3] Supryadi, **Zhang, Y.**, & Xiong, D. (2025). Targeted Neuron-Level Fine Tuning for Multilingual Toxicity Mitigation in Large Language Models. *Submitted to Transactions on Asian and Low-Resource Language Information Processing.* (*Minor Revision*).

[4] **Zhang, Y.**, Li, J., Hu, S., Zhang, T., Cui, Y., Zhu, S., & Xiong, D. (2026). SAINF: Intrinsic Self-Correction for Robust Machine Translation with Large Language Models. *Accepted by Frontiers of Computer Science (FCS) 2026.* [DOI:10.1007/s11704-026-51740-6]

## RESEARCH EXPERIENCES

**Autonomous Vehicle Trajectory Simulation and Generation Driven by LLMs** **May 2024 – Feb 2025**
*Team Leader, College Student Innovation Project / Supervisor: A.Prof. Di LIN*

- Implemented a lightweight SimCopilot variant by encoding 1,000 multi-agent T-junction scenarios into structured instruction–response pairs and LoRA fine-tuning LLaMA-7B for interaction-aware trajectory generation with <1% trainable parameters.

**Self-Adaptive Framework for Robust LLM-Based Machine Translation (SAINF)** **Dec 2024 – Apr 2025**
*Research Assistant / Supervisor: Prof. Deyi XIONG*

- Designed a self-adaptive translation framework with threshold-based conditional routing, enabling LLMs to dynamically switch between efficient Chain-of-Dictionary prompting and targeted refinement without relying on external Quality Estimation models.
- Achieved +6.9 BLEU improvement on ChallengeWMT by integrating intrinsic interpretation validation that filters low-confidence outputs before refinement.

**Neuron-Level Fine-Tuning for Multilingual Toxicity Mitigation in LLMs** **Apr 2025 – May 2025**
*Research Assistant / Supervisor: Prof. Deyi XIONG*

- Curated a multilingual instruction dataset spanning 9 languages by processing open-source hate speech corpora and generating 10,000+ non-toxic response pairs through prompt engineering and LLM-assisted annotation.
- Co-developed a targeted fine-tuning methodology that identifies and modifies toxic neurons while preserving model's general capabilities, achieving 85%+ toxicity reduction with minimal performance degradation on downstream tasks.

**GuardAlign: Robust Safety Alignment in Multimodal LLMs** **Jun 2025 – Sep 2025**
*Research Assistant / Supervisor: Dr. Junfeng FANG*

- Validated optimal transport over cosine similarity for unsafe image-patch detection ($\sim$10$\times$ larger KL on 500 SPA-VL samples), the evidence behind the final detector; the method cuts unsafe responses by up to 39% across 6 MLLMs.

**Mutual Information-Guided Reinforcement Learning for Vision-Language Models** **Oct 2025 – Apr 2026**
*Research Assistant / Supervisor: Dr. Junfeng FANG*

- Proposed a decoupled VLM training framework that leverages mutual information as a cheap pre-screening signal to enable intelligent sampling budget reallocation via forking, resolving reward blindness to visual perception quality.
- Implemented the full RL pipeline in veRL on Qwen2.5-VL-7B across 8$\times$A100 (80GB) GPUs, reaching 70.22% average accuracy across 6 benchmarks, surpassing the DAPO-16 baseline with $\sim$14% fewer computational resources.

## AWARDS & HONOURS

**Honorable Mention**, Mathematical Contest in Modeling (MCM/ICM) **Feb 2025**
**Merit Student Scholarship**, Tianjin University **2023, 2024, 2025, 2026**
**Daoda Scholarship**, Nantong Alumni Association **Oct 2022**

## LEADERSHIP & ACTIVITIES

- **ACM Europe Summer School on HPC Computer Architectures for AI and Dedicated Applications, Barcelona Supercomputing Center** (Jul 2026).
- **President, Film Research and Art Society, Tianjin University** (Sep 2023 – Sep 2024).
- **Barista & Head Instructor, 109 Coffee Society, Tianjin University** (Oct 2022 – May 2026).
- **"Most Beautiful Volunteer" Award**, Nantong City "Shunfengche" Volunteer Service.

## SKILLS

- **Languages:** Mandarin (Native), English (Fluent, IELTS 7.5).
- **Technical Skills:** Python (PyTorch, Transformers, veRL, vllm), C/C++, JavaScript, SQL, Linux, Mathematica, MATLAB, Tableau, Figma, LaTeX.